

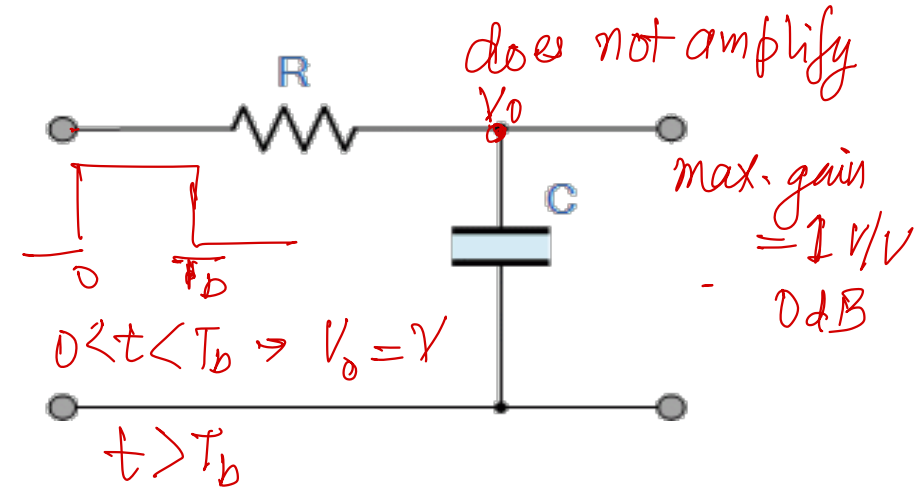
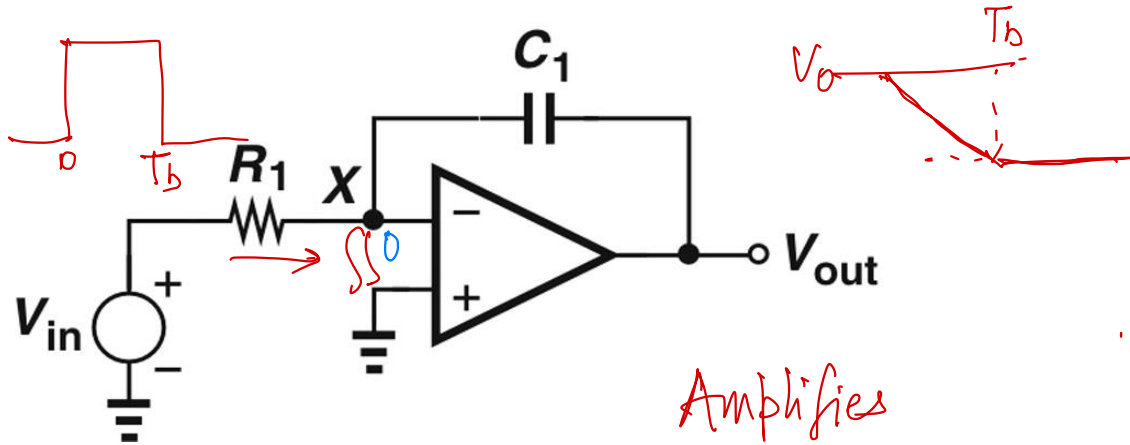
# ES -116

## Principles and Applications of Electrical Engineering

Semester II, 2025-26

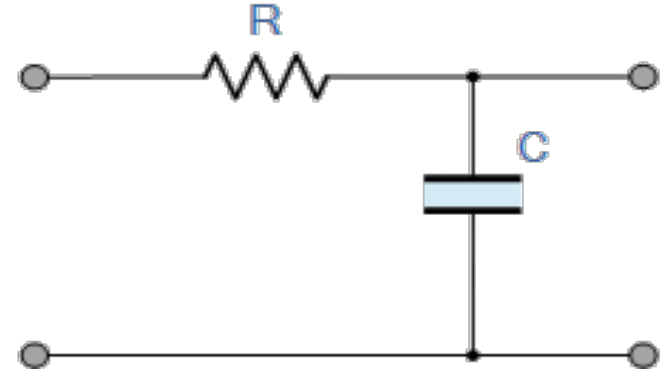
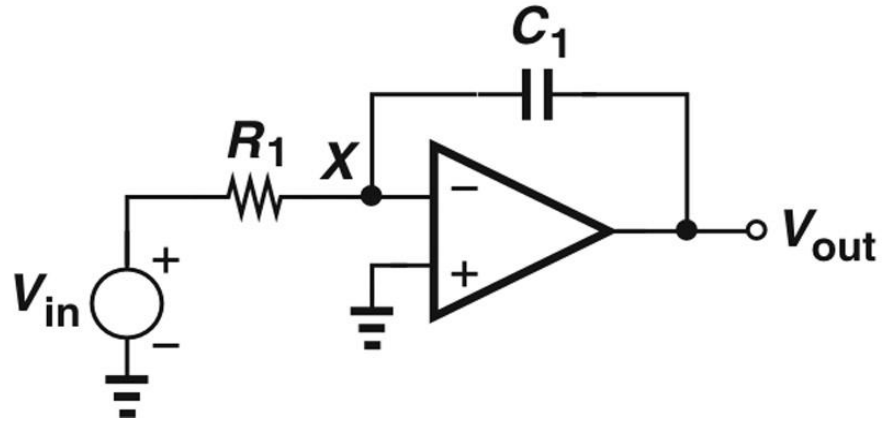
OpAMP based Functionalities (continued)

# Recap

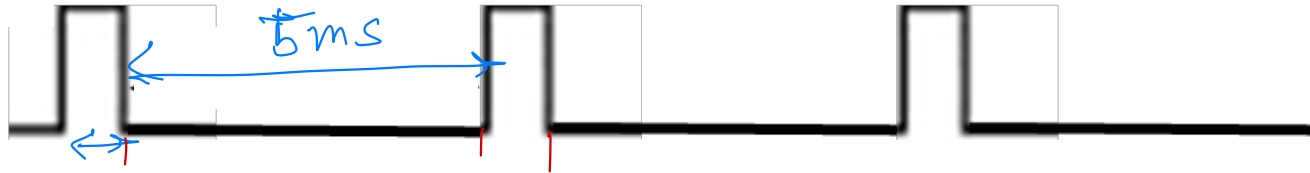
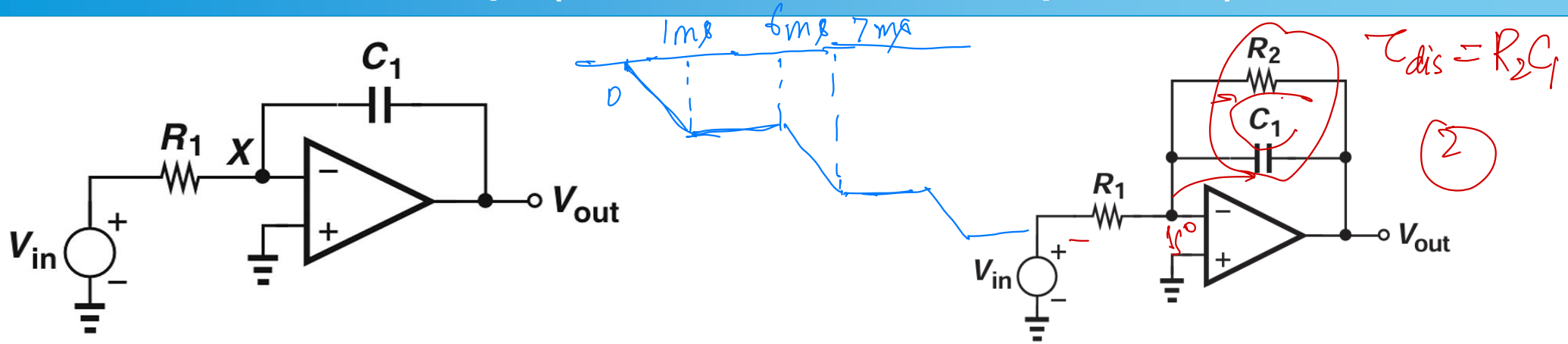


What is the difference between time domain response and frequency response of the two circuits?

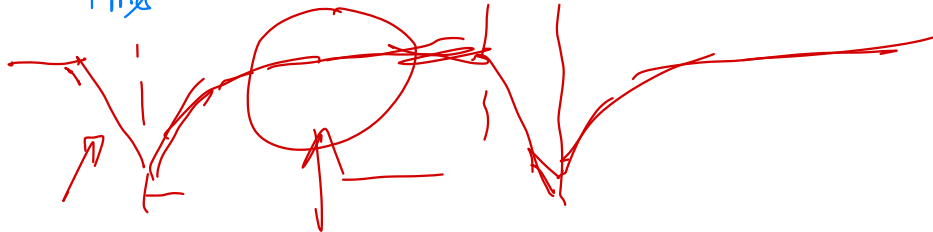
# Recap (time domain response)



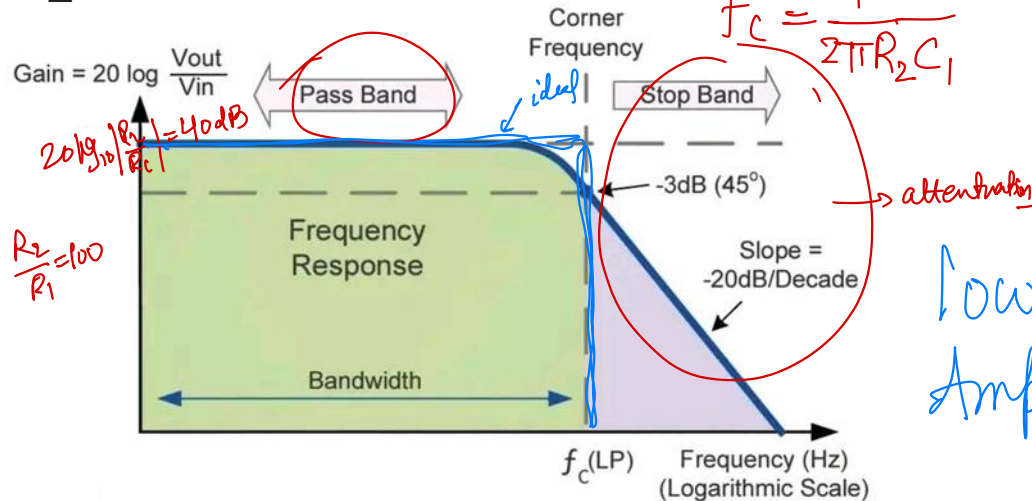
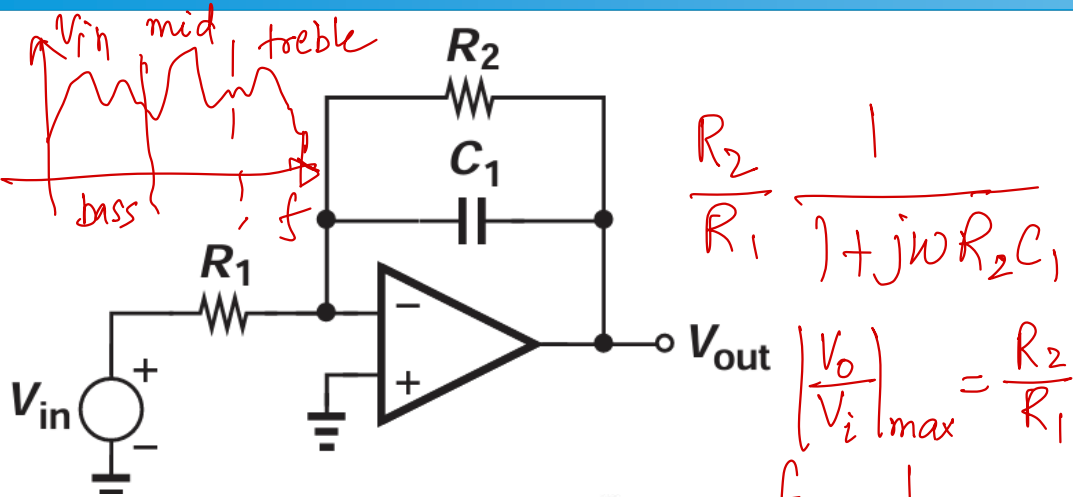
# Recap (time domain response)



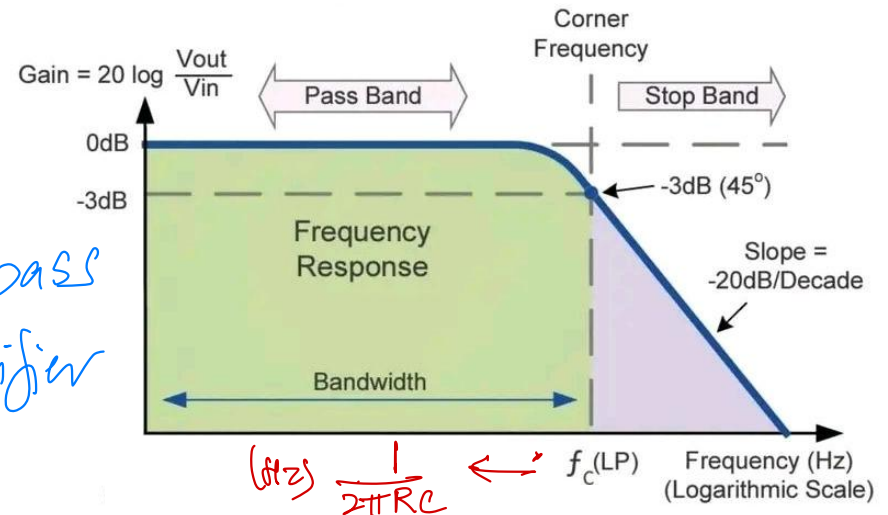
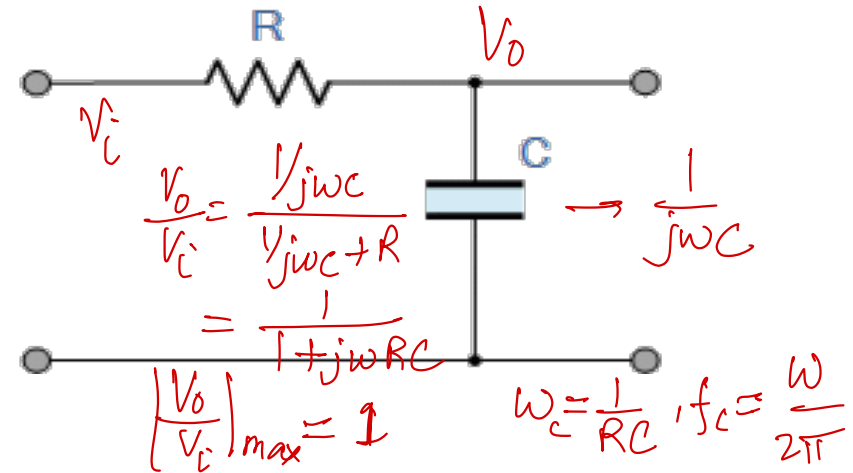
(2)



# Recap (frequency response)



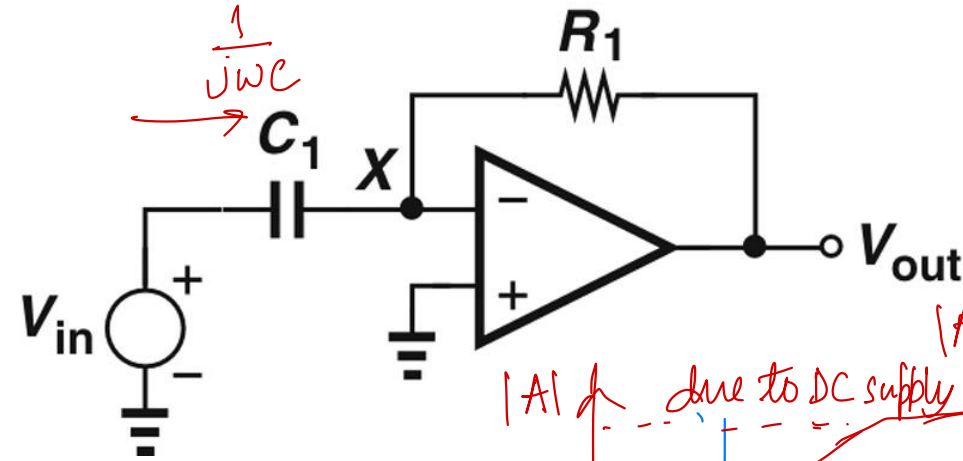
low-pass  
Amplifier



# Functionality in Frequency domain

Option 2:  $Z_1 = C_1$ ,  $Z_2 = R_1$

Plot  $V_{out}/V_{in}$  with frequency.

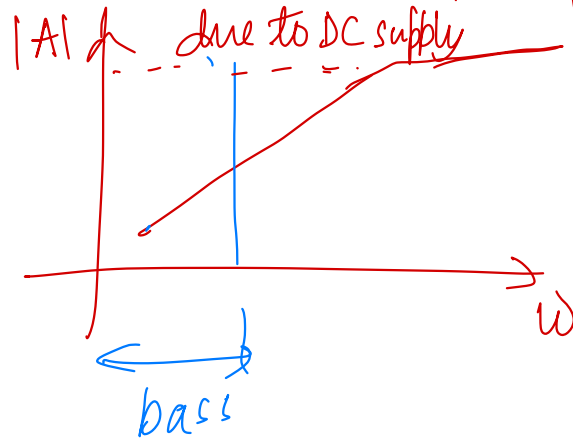


$$\frac{V_{out}}{V_{in}} = \frac{-R_1}{1/j\omega C_1} = -R_1 C_1 j\omega$$

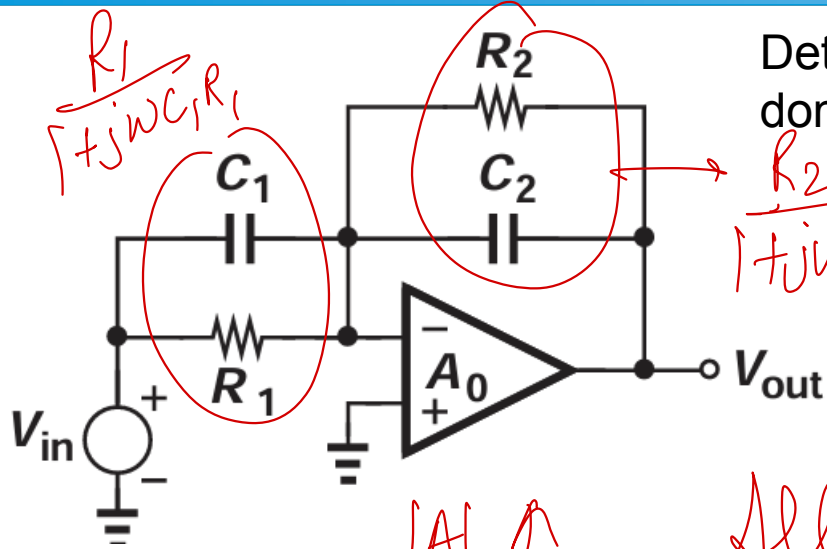
Mag      phase

$$|AF| = \left| \frac{V_{out}}{V_{in}} \right| = R_1 C_1 \omega$$

high pass amplifier



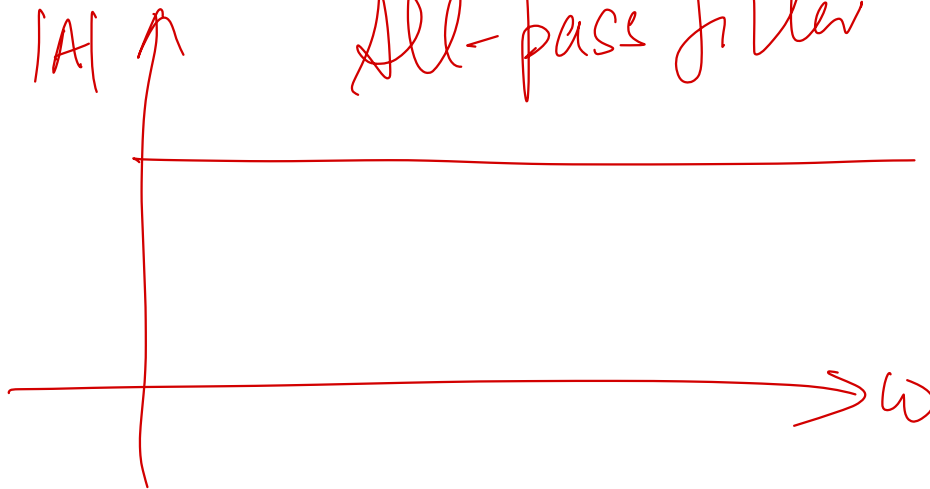
# Check your understanding



Determine the functionality of the given circuit in frequency domain (i.e. transfer function:  $V_{out}(f)/V_{in}(f)$ ).

$$A = \left| \frac{V_{out}}{V_{in}} \right| = \frac{R_2}{R_1} \sqrt{\frac{1 + (\omega C_1 R_1)^2}{1 + (\omega C_2 R_2)^2}}$$

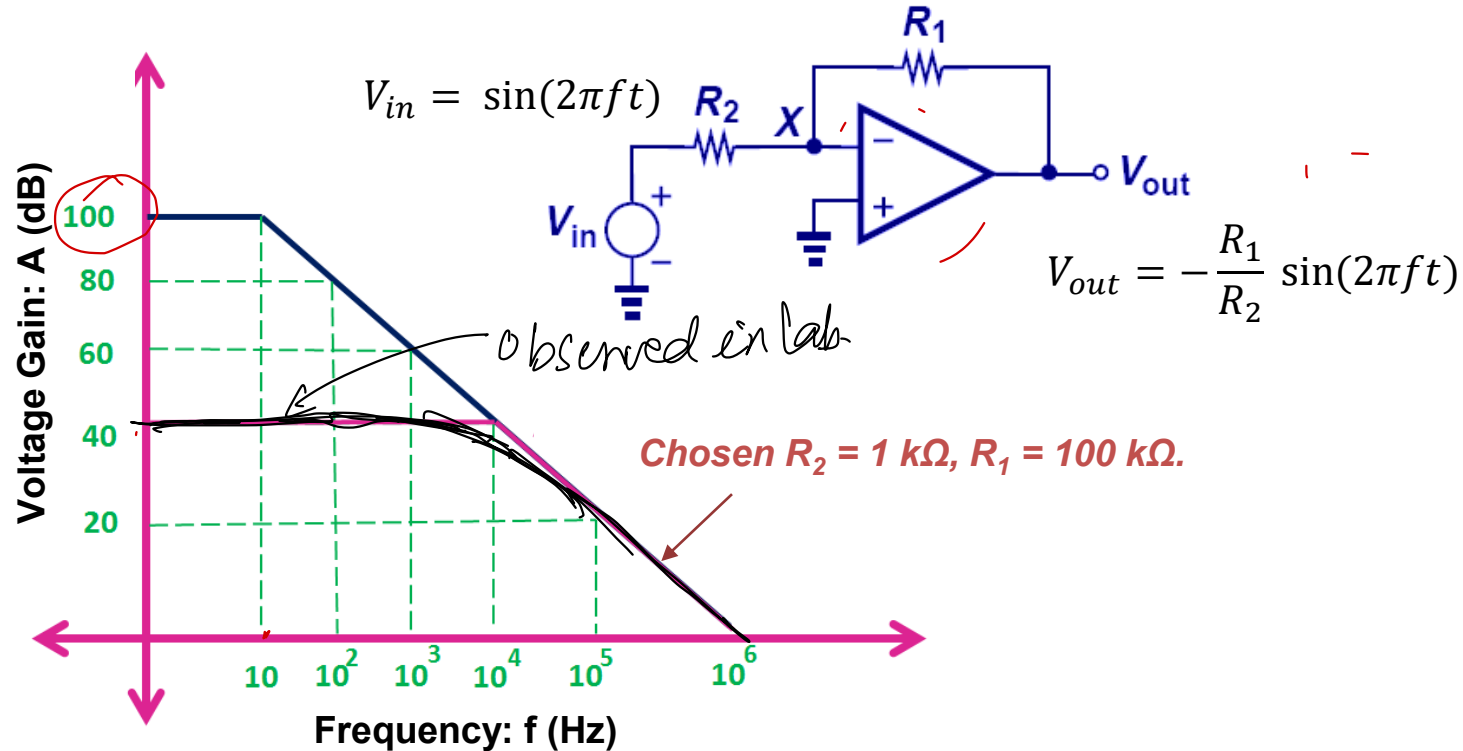
All-pass filter



$$C_1 R_1 = C_2 R_2$$

$$\omega_1 = \frac{1}{R_1 C_1} = \omega_2 = \frac{1}{R_2 C_2}$$

# How do you explain this observation?

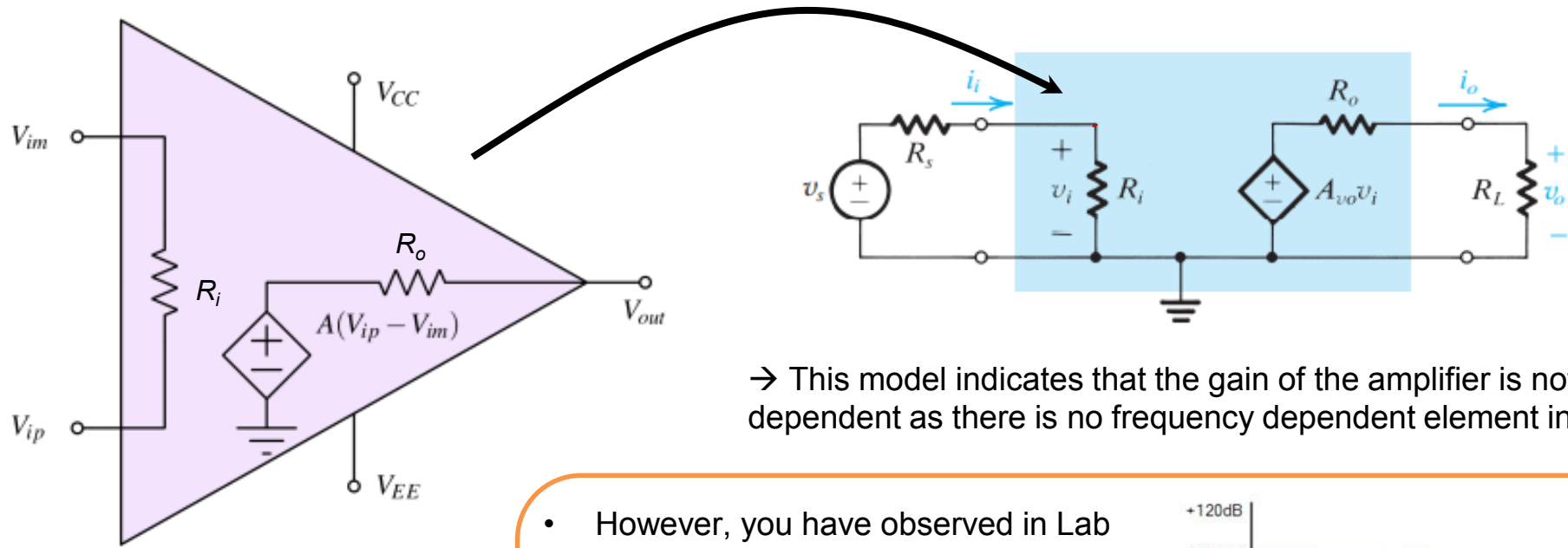


$$A \text{ (in dB)} = 20 \log_{10} \left| \frac{V_{out}}{V_{in}} \right|$$

What happens to the frequency response if we keep increasing  $R_1$  from  $100 \text{ k}\Omega$  to  $1 \text{ M}\Omega$  to  $100 \text{ M}\Omega$  (hypothetically)?

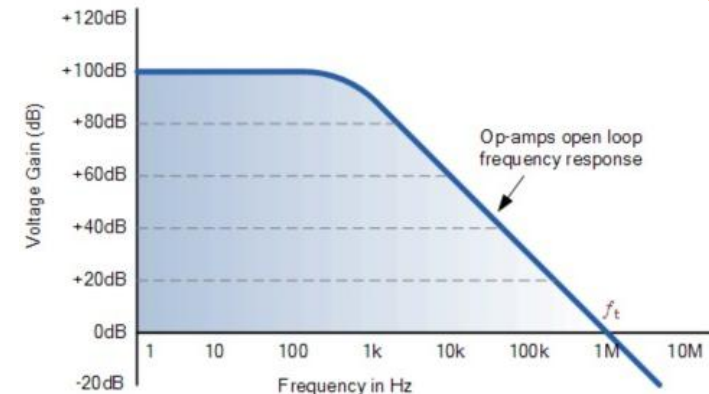


# How have described OpAMP circuit model?

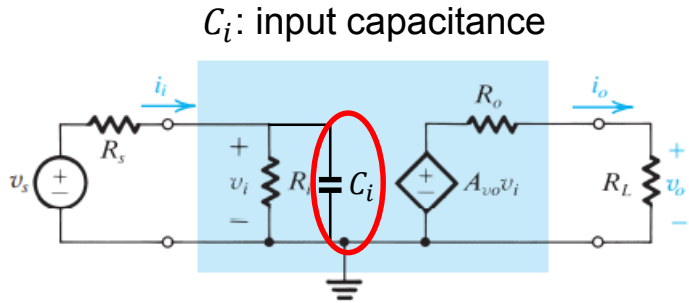


→ This model indicates that the gain of the amplifier is not frequency dependent as there is no frequency dependent element in the circuit.

- However, you have observed in Lab 7 that OpAMP based amplifier's gain decreases at higher frequencies.
- What is **missing** in the above circuit model due to which it does not capture the observed frequency dependence?



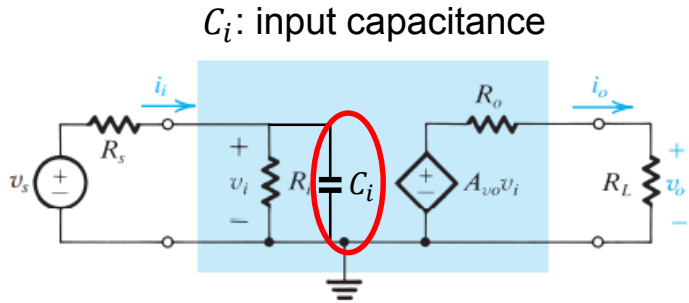
# Surprise: Frequency dependent components



Does it mean that the gain of the amplifier becomes frequency dependent?

Let's calculate  $\frac{v_o}{v_s}$

# Frequency response of an amplifier



$$\frac{v_o(s)}{v_s(s)} = A_{v0} \frac{R_L}{R_L + R_o} \frac{R_i}{R_i + R_s} \frac{1}{1 + sC_i(R_i || R_s)}$$

$$\frac{v_o(j\omega)}{v_s(j\omega)} = A_{v0} \frac{R_L}{R_L + R_o} \frac{R_i}{R_i + R_s} \frac{1}{1 + j\omega C_i(R_i || R_s)}$$

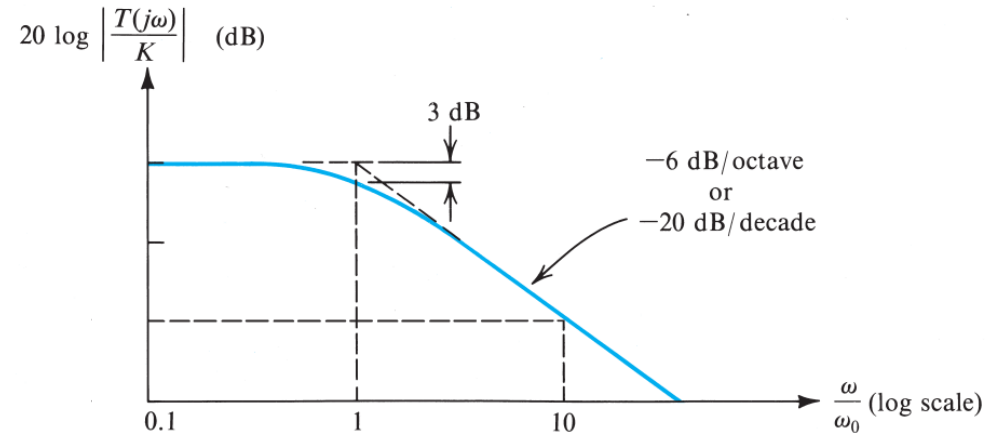
← freq. dependent term

At  $\omega = 0$ ,  $\frac{v_o(0)}{v_s(0)} = A_{v0} \frac{R_L}{R_L + R_o} \frac{R_i}{R_i + R_s} = K$  (DC gain of the amplifier as derived earlier)

Transfer function,  $T(j\omega) = \frac{v_o(j\omega)}{v_s(j\omega)} = \frac{K}{1 + j\omega C_i(R_i || R_s)}$

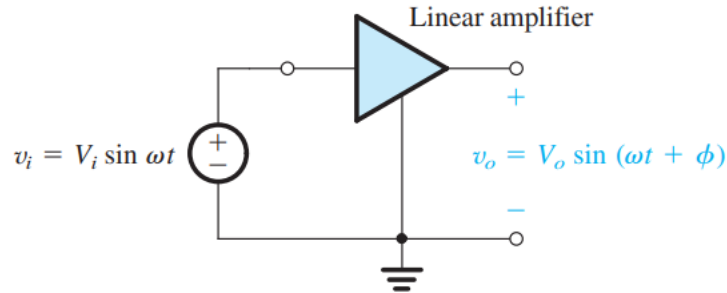
$T(j\omega) = \frac{K}{1 + j(\frac{\omega}{\omega_0})}$  where,  $\omega_0 = \frac{1}{C_i(R_i || R_s)}$  (corner frequency)

$$\left| \frac{T(j\omega)}{K} \right| = \frac{1}{\sqrt{1 + \left( \frac{\omega}{\omega_0} \right)^2}} \Rightarrow \begin{cases} \text{At } \omega = \omega_0, \left| \frac{T(j\omega)}{K} \right| = \frac{1}{\sqrt{2}} \\ 20 \log \left| \frac{T(j\omega)}{K} \right| = 20 \log \frac{1}{\sqrt{2}} = -3 \text{ dB} \end{cases}$$



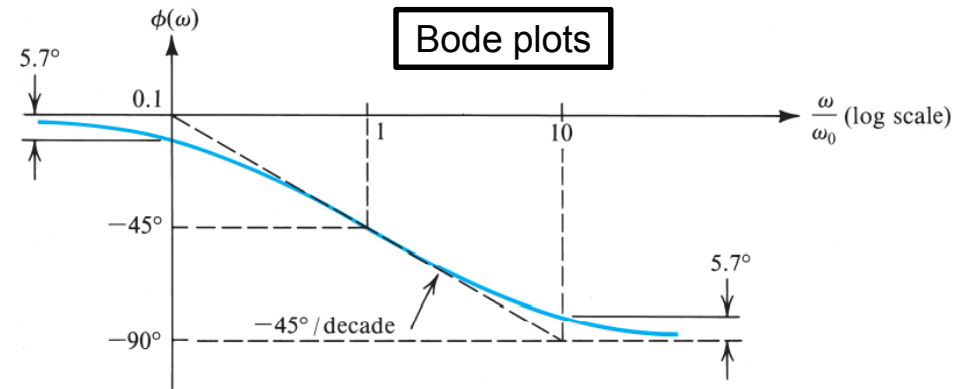
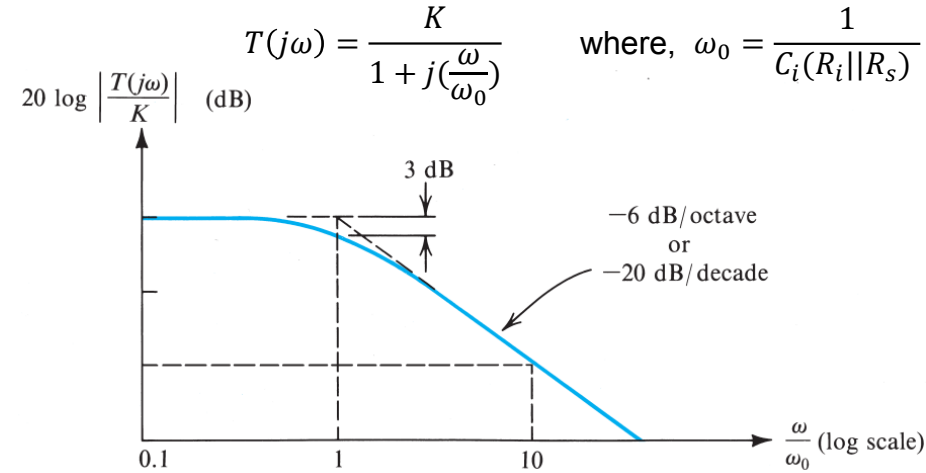
# Physical meaning of frequency response

- Input signal to amplifier can be expressed as sum of sinusoid signals.
- Frequency response: Amplifier's response to input sinusoids of different frequencies.



- By changing the frequency  $\omega$ , we can measure the magnitude of amplifier gain  $|T(j\omega)|$  and phase  $\angle T(j\omega)$  to obtain its complete frequency response.

$$\left| \frac{T(j\omega)}{K} \right| = \frac{1}{\sqrt{1 + \left( \frac{\omega}{\omega_0} \right)^2}} \quad \angle T(j\omega) = \tan^{-1} \left( \frac{\omega}{\omega_0} \right)$$

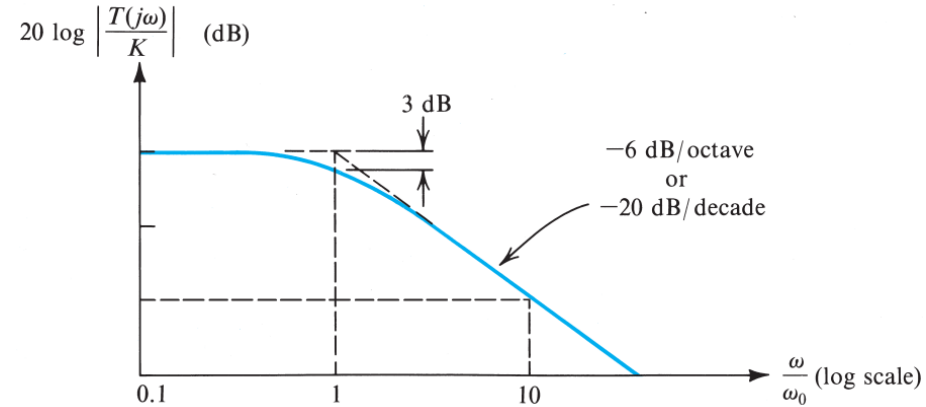


# An example (Take Home)

Suppose that you are watching Celine Dion who can sing three octaves (e.g. 216 Hz – 432 Hz – 864 Hz). During a concert, you use an amplifier with DC gain ( $K$ ) = 40 dB,  $R_i = 100\text{ k}\Omega$ ,  $C_i = 32\text{ nF}$ , and  $R_s = 10\text{ k}\Omega$ . Would this amplifier provide the same amplification for all the notes that you will hit tonight? [Hint: Calculate the 3-dB and the unity gain frequencies]

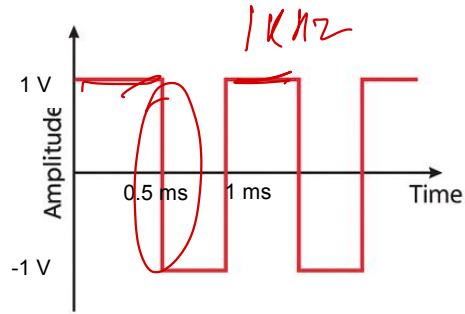
$$|T(j\omega)| = \frac{|K|}{\sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2}}$$

$$\angle T(j\omega) = \tan^{-1}\left(\frac{\omega}{\omega_0}\right) \quad \text{where,} \quad \omega_0 = \frac{1}{C_i(R_i || R_s)}$$



Key metrics of an amplifier: Gain and Bandwidth

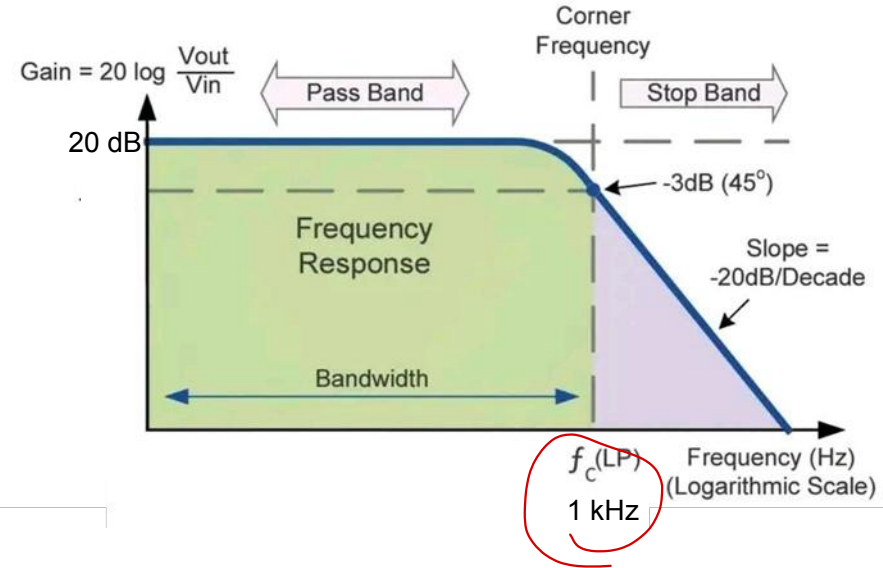
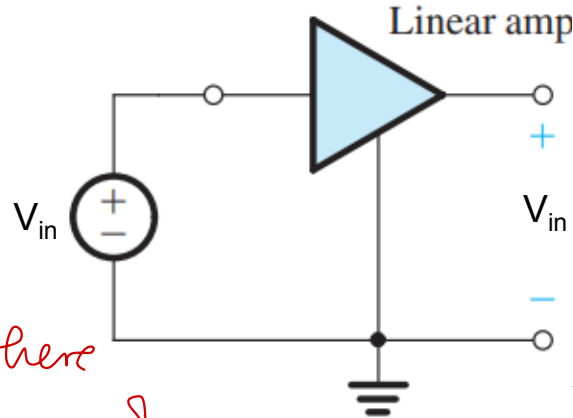
# Importance of Bandwidth in an Amplifier



How many frequencies are there in a square wave?

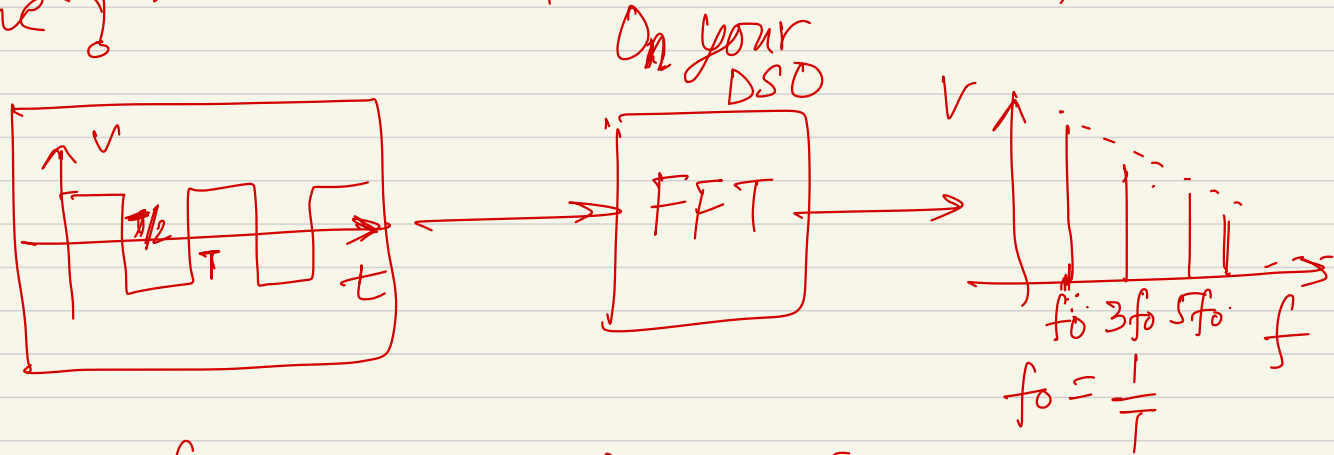
only one?  $\times$

$V$   $\uparrow$   
 $\uparrow$   
1 kHz  $\rightarrow f$



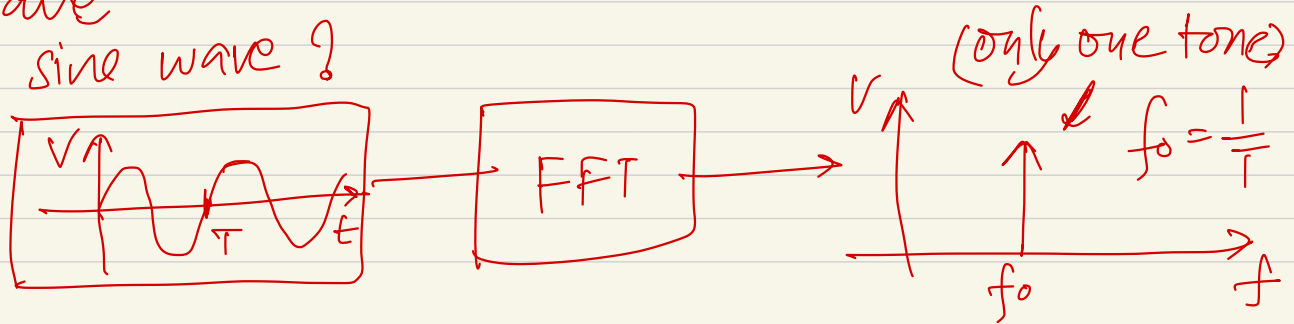
How does the output look like?

What frequencies are present in a square wave?

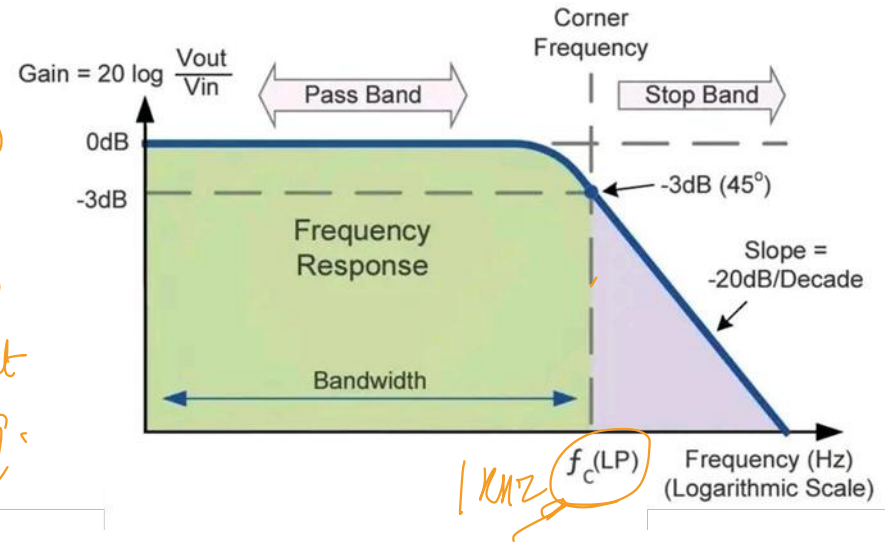
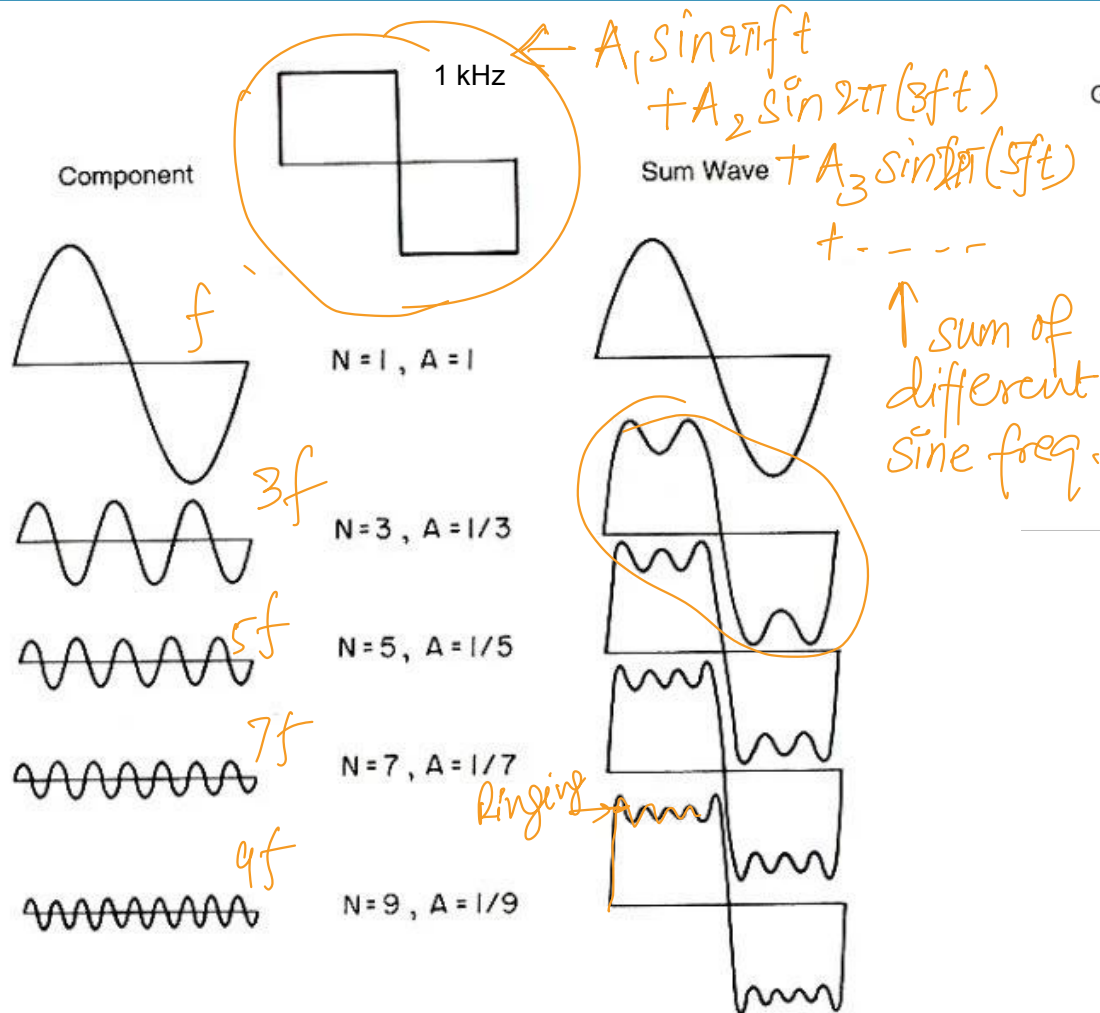


\* Multiple frequencies are present in a square wave

What about sine wave?



# What constitutes a Square wave?

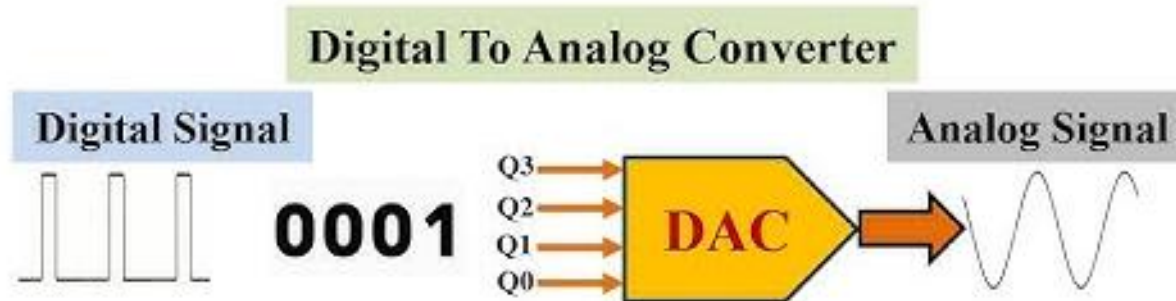
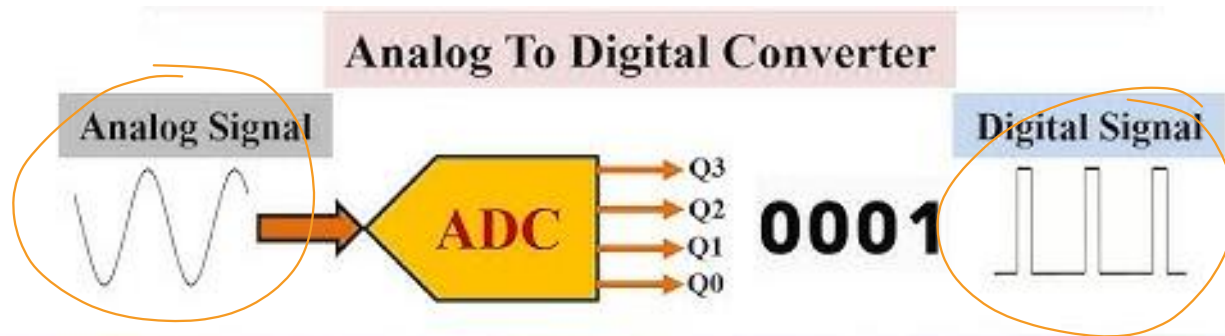


To build a reasonable square wave what bandwidth would you require?

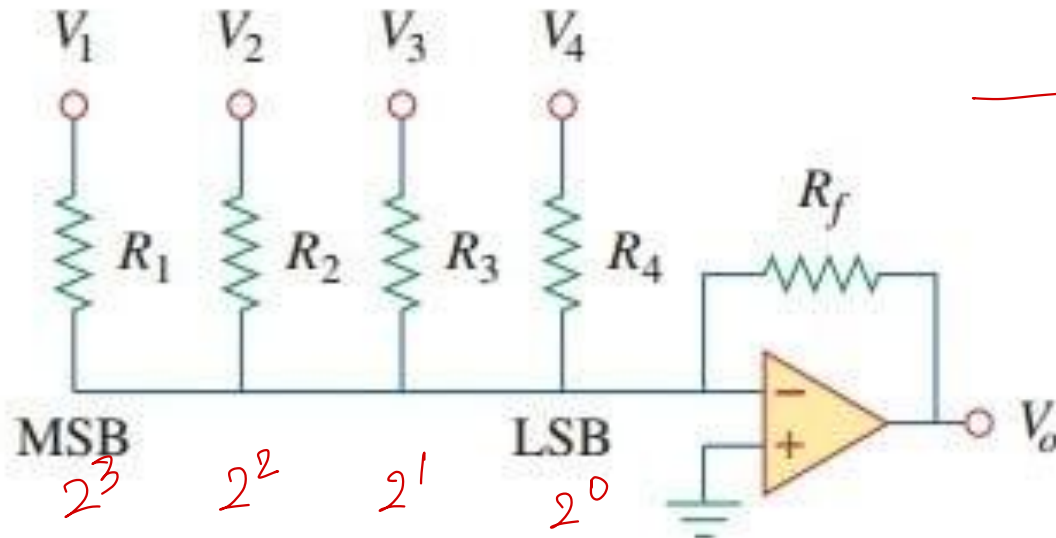
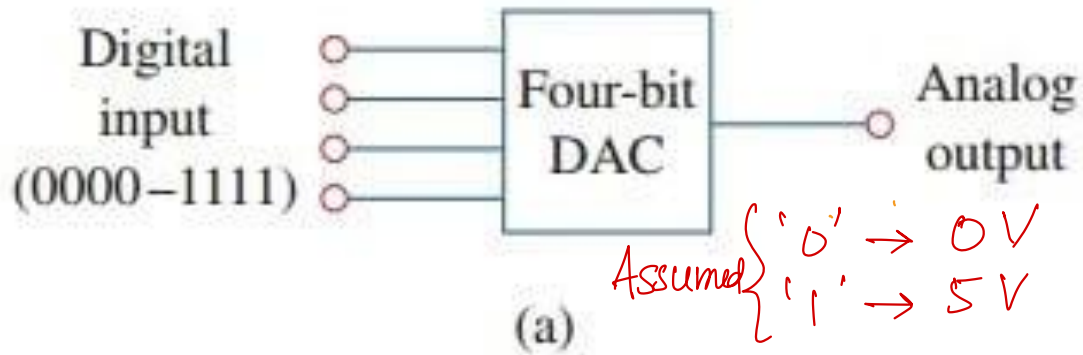
for  $f_c = 1 \text{ kHz}$ , we get a sinusoid output. therefore, for a perfect square, we require  $f_c \rightarrow \infty$ .



# Where do you find these square waves in systems?



# Let's convert a digital signal to a Analog Voltage



| $D_1$ | $D_2$ | $D_3$ | $D_4$ | $V_1$ | $V_2$ | $V_3$ | $V_4$ |
|-------|-------|-------|-------|-------|-------|-------|-------|
| 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| 0     | 0     | 0     | 1     | 0     | 0     | 0     | +5V   |
| 0     | 0     | 1     | 0     | 0     | 0     | +5V   | 0     |

1 1 1 1 + 5V + 5V + 5V + 5V

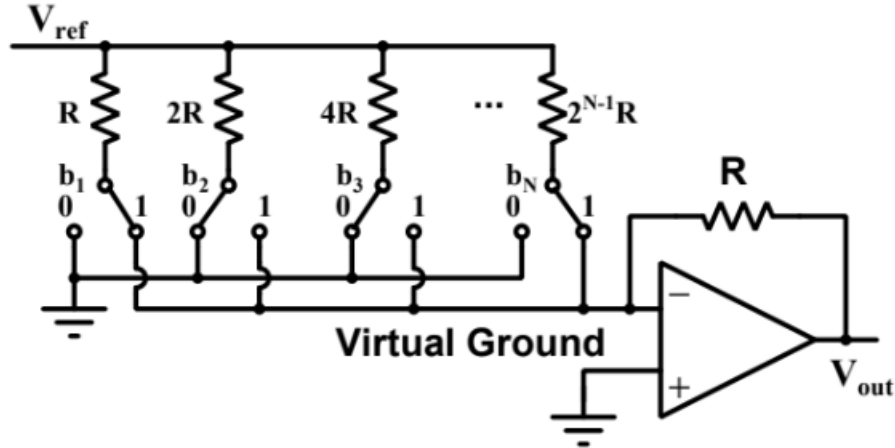
$$V_0 = - \frac{R_f}{R_1} V_1 - \frac{R_f}{R_2} V_2 - \frac{R_f}{R_3} V_3 - \frac{R_f}{R_4} V_4$$

to convert  $R_1$  Digital to Analog Form,  
we require  $2^n$  base multiplication  
 $2 = 11P$   $P = 8P$

$$\Rightarrow R_1 = R, R_2 = 2R, R_3 = 4R, R_4 = 8R$$

$$\Rightarrow V_0 = -\frac{R_f}{R} \left( V_1 + \frac{V_2}{2} + \frac{V_3}{2^2} + \frac{V_4}{2^3} \right)$$

# Binary Weighted Resistors



- Each bit corresponds to a switch.
- Current goes to ground if bit is 0.
- Currents are added into summing node if bits are 1.
- The opamp converts current back to voltage.

$$V_{out} = -R \sum_{i=1}^N \frac{V_{ref}}{2^i R} b_i = -\frac{V_{ref}}{2^N} \sum_{i=1}^N 2^{N-i} b_i$$

What is the downside of this circuit? [Hint: Does it scale well with number of bits]

# Another implementation (Lab 10)

